

Experiment 12 :

Implement an Earley parser for context-free grammars using a simple python program.

Code:

```
import nltk

from nltk import CFG
from nltk.parse import EarleyChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the'
N -> 'cat' | 'mouse'
V -> 'chased'
""")

parser = EarleyChartParser(grammar)

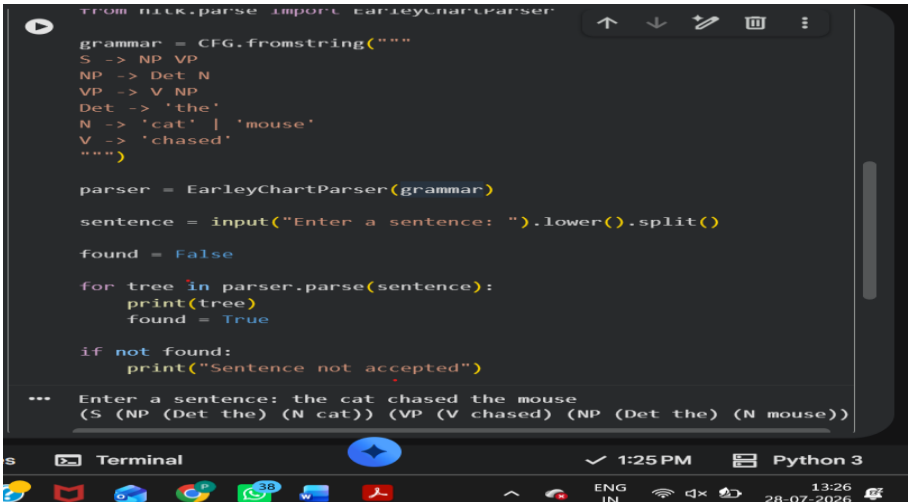
sentence = input("Enter a sentence: ").lower().split()

found = False

for tree in parser.parse(sentence):
    print(tree)
    found = True

if not found:
    print("Sentence not accepted")
```

Output



```
from nltk.parse import EarleyChartParser

grammar = CFG.fromstring("""
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*** Enter a sentence: the cat chased the mouse
(S (NP (Det the) (N cat)) (VP (V chased) (NP (Det the) (N mouse)))
```